Forging the Future: Market Intelligence and Growth Strategy for High-Performance Metals Through 2035

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1. Introduction

For centuries, metals have been among the essential materials on which human societies are built, enabling the construction of cities, transport systems, power networks, machinery and technologies that support modern economies. Their uses have expanded with each wave of industrial development, and today they remain central to established industries as well as emerging technologies. This enduring role is reflected in the sector's economic scale: according to McKinsey & Company's Global Materials Perspective, the global metals and mining industry generated approximately US$3.0 trillion in revenue in 2024, after a decline of 6 percent, and grew by only about 1 percent in 2025 (McKinsey & Company, 2025, 2026).

Within this broad market, high-performance alloys serve applications where ordinary materials may not meet the required operating conditions. These engineered materials combine metals such as nickel, cobalt, titanium, iron and chromium to provide properties such as strength at high temperatures, corrosion resistance, oxidation resistance or reliable performance under pressure. Nickel- and cobalt-based superalloys, titanium alloys and corrosion-resistant specialty steels are examples. Demand comes from aerospace and defence, power generation, chemical processing, oil and gas, medical technology and other industries with demanding technical requirements. The production and availability of metals including nickel, cobalt, chromium, molybdenum, titanium, tungsten, zirconium, copper, aluminium and manganese therefore provide important context for understanding this specialist market.

The outlook for high-performance alloys to 2035 is positive, although growth is likely to differ across alloy types and applications. One market estimate places the segment at US$13.6 billion in 2025 and projects it to reach US$19.0 billion by 2034 (The Insight Partners, 2025), although other providers start as low as US$10.2 billion because they draw the boundary of the segment differently (IMARC Group, 2025), while broader industry analysis points to continued demand from aerospace, defence, energy and other industrial uses. This growth will depend on the pace of investment and technology adoption, as well as the cost and availability of alloying metals. The strongest prospects are likely to be in applications where a particular combination of material properties, reliability and qualification requirements creates lasting value for customers.

2. Report Objective and Scope

This project studies selected metals that are essential to high performance alloys and follows them from the mine to the industrial applications that depend on them, in order to assess how their markets may develop through 2035. It asks one central question: how do the concentration of metal supply, the conversion of metals into qualified product forms and the needs of seven industries determine where the most attractive opportunities in high

performance materials will lie through 2035? Four subsidiary questions structure the answer, and each maps onto one part of the report. The first asks where the ten metals are mined and refined, how exposed that supply is to concentration and export policy, and in which qualified forms the metals reach industry. The second asks which industries and regions drive demand for these forms today and how large the measurable market is. The third asks how large demand could become by 2035, how certain that estimate is and which emerging applications could add to it, and the fourth asks where supply security, growth and entry barriers combine into the most attractive opportunities.

The study covers ten metals: nickel, cobalt, chromium, molybdenum, titanium, tungsten, zirconium, copper, aluminium, and manganese. It examines seven industries: automotive, oil and gas, energy and environmental technology, aerospace, chemical processing, electronics and electrical engineering, and medical technology. The study uses the latest official data available in November 2026. It covers the global market and compares Europe, North America, and Asia. The outlook quantifies established uses and evaluates emerging uses separately because official statistics do not yet capture them.

The report argues that much of the value in this market comes from turning metals into qualified products. Nickel is in surplus, while export policies restrict supplies of cobalt, tungsten, and molybdenum. Countries that make qualified bar, strip, and tube earn trade surpluses on those products that mining countries do not.

Measured in tonnes, the largest growing pool is superalloy bar, billet and remelt stock for engines and turbines. It rises from about 193 kt in 2025 to 248 kt in 2035, worth EUR 6.3 to 11.7 bn (Product Strategy 2035 Team, 2026a). By index, electronics grows fastest at 141, ahead of aerospace and medical technology at 136, energy at 132 and automotive at 102 (Product Strategy 2035 Team, 2026a). Electronics tracks fab capacity with unmeasured alloy volume, so it yields a priority 2 position in fab gas strip and tube. Priority 1 goes to turbine remelt stock and sheet and to Alloy 718 engine bar, both at 3.85, for buyers in the United States and Europe. As one vacuum melt programme, they reach priority 1 under three of four weightings (Product Strategy 2035 Team, 2026a).

The ranking is conditional, since a tighter Congo cobalt quota or a severe aerospace downturn each moves one of them to priority 2 (Product Strategy 2035 Team, 2026a). Priority 2 covers engine bar in all grades, CoCr implant bar, Alloy 690 and additive powder, while FeNi36 mask foil and NO96 strip remain options. Producers should stop investing in diesel only parts, new European base chemical plants, unfunded electrolyser capacity and captive zirconium cladding.

3. Introduction to the Analytical Framework and Methodology

The study follows the seven links of the metal value chain: material, product form, application, industry, geography, market size, and growth. Table 1 shows which chapters cover each link. Figures are compared only at the same stage, so contained metal is kept apart from gross weight and output from capacity. Mine output, alloy melt, mill products, and industry revenue are reported separately.

Structured data from the USGS, UN Comtrade, Eurostat PRODCOM, the World Bank, OECD, UNIDO, and the IEA form the base. Associations, official outlooks, and 61 company profiles add detail, and commercial reports serve only as a cross-check. Key figures are checked against a second source where one exists. The source register records each value's unit, stage, type, source, and formula and flags single-source figures.

Table 1: The metal value chain and its evidence

Link	Guiding question	Main evidence	Chapter
Material	How concentrated is supply?	USGS, IEA, World Bank prices	4
Product form	In which qualified forms does it reach industry?	PRODCOM, producer filings	4
Application	Which components need it, and why?	Standards, associations, filings	4 and 5
Industry	Which industries buy, and how fast do they grow?	Airbus, Boeing, IEA, OICA, Cefic, SEMI	5
Geography	Where are the forms made and bought?	IEA regions, OECD, UN Comtrade	5 and 6
Market size	How large is measurable demand today?	Apparent consumption, trade values, family tonnes	5 and 6
Growth	How large could demand be in 2035?	Demand model, emerging options register	7

Tier 1 estimates established demand as industry activity times metal content per unit times the share of units using a high-performance alloy. Activity measures such as engine deliveries, gas turbine gigawatts, and joint replacements start from measured 2025 values. To test the model, the team fitted trends to data through 2019 and compared them with 2020 to 2025 outcomes and with earlier official outlooks. A Monte Carlo simulation gives the 10th, 50th, and 90th percentiles of 2035 demand, and a supply analysis tests export quotas and licensing changes.

Tier 2 records emerging applications that current data cannot yet confirm. Examples are hydrogen-fueled turbines, supercritical CO2 recuperators, molten salt and small modular reactors, fusion magnet tape, and printed engine parts, drawn from patents, pilot plants, draft

standards, and announced projects. Each is rated for technology readiness, evidence, alloys and forms, and earliest commercial volume.

The register names the event, such as a final investment decision or first serial order, that would move each candidate into tier 1. Chapter 8 scores both tiers on the same eight criteria, so producers can see which opportunities to monitor and which merit funding.

4. Metal Markets: Materials, Product Forms and Applications

The value chain starts with the metal. Table 2 shows the commercial forms of the ten metals in this chapter and their main uses.

Table 2: Commercial forms and main uses by metal

Metal	Commercial forms	Main use and company example	Share
Nickel	Bar Billet Sheet Plate Tube Battery salts	Stainless steel (Alfa Laval T10 316L plate) Batteries (LG Energy Solution M50L cell) Nickel alloys (Haynes C-276 reactor plate)	66% 16% 6%
Cobalt	Remelt stock Bar Powder Cobalt salts	Batteries (LG Energy Solution M50L cell) Superalloys (Haynes 188 combustor sheet)	≈75% ≈7%
Chromium	Ferrochrome Chromium metal Sheet Plate	Stainless steel (Franke MYX 110-55 sink)	>80%
Molybdenum	Ferromolybdenum Plate Sheet Tube	Chemicals & petrochemicals (Alfa Laval T10 C-276 plate) Oil & gas (Alleima SAF 2507 umbilical tube)	17% 14%
Titanium	Sponge Plate Bar Tube Forgings	Aerospace (Airbus A350 landing gear)	>50%*
Tungsten	Carbide powder Inserts Tungsten metal	Carbide products (Sandvik CNMG 4425 insert) Steels & superalloys (Haynes 188 combustor sheet)	65% 14%

Metal	Commercial forms	Main use and company example	Share
Zirconium	Zircon concentrate Sponge Plate Tube	Ceramics (Iluka Premium zircon for tile) Acid service (Alleima Zr 702 exchanger tube) Fuel cladding (Framatome M5 fuel rod cladding)	>50% n.r. n.r.
Copper	Wire Tube Sheet Strip Bar	Construction (Southwire SIMpull THHN wire) Consumer, cooling & electronics (Wieland grooved tube) Power & telecom (Prysmian 12/20 kV Cu cable)	26% 23% 17%
Aluminium	Sheet Plate Extrusions Castings Conductor rod	Construction (WICONA WICLINE 75 Eco frame) Transport (Novelis s118 hood inner sheet) Electrical (Southwire Drake ACSR conductor)	≈25% ≈23% ≈12%
Manganese	Ferromanganese Silicomanganese Electrolytic metal	Steelmaking (voestalpine MN13 rail crossing)	≈97%

*Examples name a supplied component or material. Shares cover the full use category. First use: Ni (2024) and W (2021). Demand or end use: Co and Mo (2025), Cu (2024), Al (2020), Mn (undated). Cr covers ferrochrome and Zr covers zircon output. *Ti covers U.S. metal use (2025), reported as a majority. n.r. = not reported. Sources: (Cobalt Institute, 2026; CRU International, 2022; Iluka Resources, 2025; International Chromium Development Association, 2022; International Copper Study Group, 2025; International Manganese Institute, n.d.; International Molybdenum Association, 2026b; International Tungsten Industry Association, n.d.; Nickel Institute, n.d.; U.S. Geological Survey, 2026i).*

Ferroalloys go into the melt, while plate, tube, bar and powder go to component makers. The profiles below show how these forms are supplied.

4.1 Nickel

Indonesia has become the largest source of nickel within a decade. The USGS estimates 2025 mine output at 3.90 Mt of contained nickel, with 2.60 Mt, or about two thirds, from Indonesia. The IEA estimates 4.21 Mt, with Indonesia at 63 percent of mining and 45 percent of refining, so both show concentrated supply (International Energy Agency, 2026c; U.S. Geological Survey, 2026g).

Primary nickel output reached 3.875 Mt against 3.577 Mt of use, and just over half of Chinese and Indonesian supply was nickel pig iron for stainless steel (International Nickel Study Group, 2025, 2026). Vacuum-melted alloys need class 1 nickel, which is widely available, because the Chinese share of open LME tonnage rose from 47 to about 68 percent in the first nine months of 2025. The real constraint is approved feedstock from sources such as Vale in Canada and Glencore Nikkelverk in Norway (International Nickel Study Group, 2025; United Nations Statistics Division, 2026a). US and UK sanctions bar Russian metal produced from 13 April 2024 from US imports and exchange warrants (U.S. Department of the Treasury, 2024).

The surplus explains the falling price, from a World Bank average of USD 25,834 per tonne in 2022 to USD 15,162 in 2025 (World Bank, 2026a). Indonesia first cut its 2026 ore quota to 260 to 270 Mt, then raised it to 290 to 300 Mt, and the International Nickel Study Group cut its 2026 surplus forecast to 19 kt, after 299 kt in 2025. The quota is therefore the main swing factor for price (Argus Media, 2026a; Indonesia Business Post, 2026; International Nickel Study Group, 2026).

Figure 1: The nickel price has fallen by about 41 percent from its 2022 annual average

Source: World Bank Pink Sheet, annual averages of monthly prices, team calculation (World Bank, 2026a, 2026b).

Stainless steel uses most nickel, with melt shop output of 64.2 Mt gross in 2025. The IEA expects nickel demand outside energy technologies, including stainless steel and alloys, to rise from 2,906 kt in 2025 to 3,255 kt in 2035 (International Energy Agency, 2026c; worldstainless, 2026).

Nickel alloys reach industry as strip, sheet, plate, bar, wire, and tube. In 2024, Germany exported 17,775 t of nickel plate, sheet, and strip worth USD 578 million, about USD 32,500 per tonne or 1.9 times the nickel price (Eurostat, 2026a; United Nations Statistics Division, 2026d). That ratio overstates conversion value, because alloys contain other metals. Alloy C-276, with about 57 percent nickel, 16 percent molybdenum, and 16 percent chromium, held metal worth about USD 19,000 per tonne before tungsten. The team estimates the export unit value at about 1.7 times that metal content (Haynes International, 2023b; U.S. Geological Survey, 2026c, 2026f; World Bank, 2026a).

Producers pass metal costs on through surcharges or index clauses, as ATI does for most of its 70 million pounds of nickel, and Carpenter for sales outside the 43 percent under firm prices (ATI, 2026a; Carpenter Technology, 2026). Margins therefore depend on base prices. Special Metals raised its Huntington base prices eight times from April 2021 to February 2024, by 5 to 20 percent each time (Special Metals Corporation, 2024). Form and qualification explain why buyers pay above metal value, while cobalt adds a sharper supply risk.

4.2 Cobalt

Export restrictions and concentrated mining make cobalt scarce. The USGS estimates 2025 mine output at 310 kt of contained cobalt. The Democratic Republic of the Congo mined about 230 kt, which the USGS puts at 73 percent of the world total. China refined 78.6 percent of the world's 240 kt of refined cobalt (Cobalt Institute, 2026; U.S. Geological Survey, 2026d).

In February 2025, Congo banned cobalt exports, then set an annual quota of up to 96,600 t for 2026 and 2027. CMOC mined 117,549 t in 2025 and received an export allowance of 31,200 t. European cobalt metal prices rose 130 percent in 2025 (Cobalt Institute, 2026; Mining Weekly, 2026; Mining.com, 2026; U.S. Geological Survey, 2026d). The IEA expects 2035 mine supply of 225 kt against demand of 300 to 322 kt, a lasting gap (International Energy Agency, 2026c).

Superalloys use a small share of cobalt but need high-purity metal. In 2025, batteries used 206 kt and superalloys 20 kt. A few Western refineries supply most of the superalloy grade (Cobalt Institute, 2026). Its uses in turbine remelt stock, cobalt-chromium implants, and wear-resistant alloys are hard to switch, so a quota set in Kinshasa can disrupt production in Toulouse and Cincinnati.

4.3 Chromium

Chromium gives corrosion-resistant and heat-resistant alloys their protective oxide layer, but it is mined and smelted in different countries. World chromite ore output reached an estimated 51 Mt gross weight in 2025. The USGS puts South Africa at about 23 Mt, while the International Chromium Development Association reports 25.2 Mt on a different definition (International Chromium Development Association, n.d.; U.S. Geological Survey, 2026c). China has little chromite but smelts 9.9 Mt of ferrochrome, mostly for stainless steel. The United States imports 79 percent of its chromium needs (International Chromium Development Association, n.d.; U.S. Geological Survey, 2026c; worldstainless, 2026).

Vacuum-melted nickel alloys need purer inputs, such as low-carbon ferrochrome and aluminothermic chromium metal. The US chromium metal price rose to an estimated USD 5.90 per pound in 2025, while ferrochrome fell (U.S. Geological Survey, 2026c). AMG Chrome in the United Kingdom supplies low-gas chromium metal for single-crystal turbine blades. A vacuum melt shop can stop if it loses this qualified source (AMG Critical Materials N.V., n.d.).

4.4 Molybdenum

Molybdenum helps alloys resist hot acids and seawater, in amounts that depend on the grade. One tonne of 316L stainless steel contains about 21 kg, but Alloy C-276 holds about 160 kg and Alloy 625 about 90 kg (ATI, n.d.; Haynes International, 2023b; Outokumpu, 2025).

Much of the world's molybdenum is a by-product of copper mining, so supply responds slowly to price. The USGS estimates 2025 mine output at 260 kt, while the International Molybdenum Association reports 672.6 million pounds, about 305 kt, with almost all of the gap in China (International Molybdenum Association, 2026a; U.S. Geological Survey, 2026f).

According to the USGS, only China and the United States produce from primary molybdenum mines. Since February 2025, China has required licences for molybdenum powder exports. The oxide and ferromolybdenum that melt shops buy are excluded (Ministry of Commerce of the People's Republic of China, 2025a; U.S. Geological Survey, 2026f). The US oxide price rose 7 percent to about USD 51 per kg in 2025, while European use fell 7 percent with the chemical slowdown (International Molybdenum Association, 2026a; U.S. Geological Survey, 2026f).

4.5 Titanium

Titanium replaces nickel alloys in lighter and cooler aerospace parts. In 2025, world ilmenite and rutile output reached 9.8 Mt as TiO2, mostly for pigment, but titanium sponge output was only 370 kt (U.S. Geological Survey, 2026i, 2026j).

China produced about 260 kt of sponge and held 320 kt of the world's 470 kt of capacity. Western aircraft engines use other qualified sources, and the United States has no operating sponge plant (U.S. Geological Survey, 2026i). Russia produced about 25 kt of sponge in 2025 and shipped USD 715 million of titanium to the ten reporters in 2024, and its VSMPO-AVISMA calls itself the leading titanium supplier for aviation (U.S. Geological Survey, 2026i; United Nations Statistics Division, 2026a; VSMPO-AVISMA Corporation, n.d.).

The scarce forms are aircraft-grade sponge and the ingot, bar, and plate made from it. Osaka Titanium is raising aircraft-grade capacity from 40 kt to 50 kt by fiscal 2027. Kobe Steel sells titanium castings and forgings for aircraft, and ATI reports lead times of about 20 months for premium titanium. Europe makes no sponge but sold 22.6 kt of wrought titanium in 2024 and imported 31.7 kt (ATI, 2026b; Eurostat, 2024; Kobe Steel Ltd., 2026; Osaka Titanium Technologies, 2025). Nickel superalloys face the same shortage of qualified melting capacity.

4.6 Tungsten

China controls most of the world's tungsten supply. In 2025, mines produced an estimated 85 kt of contained tungsten, of which China produced about 67 kt, or 79 percent (U.S. Geological Survey, 2026k). China's first 2025 mining quota covered 58,000 t of concentrate. Since February 2025, it has required export licences for ammonium paratungstate, tungsten oxides, unsintered carbide, and specified tungsten alloys. Sintered carbide tools and tungsten wire remain outside these controls (Ministry of Commerce of the People's Republic of China, 2025a, 2025b; Ministry of Natural Resources, 2025).

Carbide for cutting and mining tools takes 65 percent of first use, and steels and superalloys take 14 percent (International Tungsten Industry Association, n.d.). In superalloys, tungsten strengthens cast turbine parts and heat-resistant sheet, so export licences can affect gas turbine output. Sandvik's powder business more than doubled orders and revenue in the third quarter of 2026 as tungsten prices rose (Sandvik, 2026).

4.7 Other Selected Metals

Zirconium and the minor superalloy elements serve narrow but demanding uses, while copper, aluminium, and manganese have large markets but small high-performance roles.

4.7.1 Zirconium

World output of zirconium concentrates reached about 1.2 Mt gross weight in 2025, led by Australia with 400 kt. Most goes into ceramics, foundry sand, and refractories, and the metal market is far smaller (U.S. Geological Survey, 2026l). Fuel makers and integrated producers such as ATI control their own nuclear cladding supply, so the open market covers mainly plate and tube for acid service (ATI, 2026a).

4.7.2 Copper

Copper links high-performance materials to electrification, with 2025 mine production of about 23.2 Mt. Flat-rolled products take 11 percent of first use, against 63 percent for wire. Copper averaged USD 9,947 per tonne in 2025 and set records above USD 14,500 in 2026 (International Copper Study Group, 2025, 2026; International Energy Agency, 2026c; The Oregon Group, 2026; World Bank, 2026b). Under current policies, the IEA expects output from current mines to fall to 20.7 Mt by 2035, while demand rises from 27.8 Mt to 32.4 Mt. That implies a primary supply deficit of about 25 percent (International Energy Agency, 2026b, 2026c). Specialized forms include copper-nickel and high-conductivity strip, such as the busbars in section 5.6.

4.7.3 Aluminium and Manganese

Aluminium can replace heavier alloys where temperatures and corrosion allow. World primary output reached about 74 Mt in 2025, 61 percent of it in China. European primary output has fallen to about 955 kt, against demand of about 9 Mt (European Aluminium, n.d.; Federation of Aluminium Consumers in Europe, 2024; International Aluminium Institute, 2026; U.S. Geological Survey, 2026b).

Manganese demand follows carbon steel output. Mines produced about 20 Mt of contained manganese in 2025, about 81 percent from South Africa, Gabon, Ghana, and Australia. The ore price fell to about USD 4.5 per metric ton unit, so supply risk and growth potential for

alloy producers are limited (International Manganese Institute, 2026; U.S. Geological Survey, 2026e).

4.7.4 Additional Metals

Iron matters in FeNi36 for display masks and FeNi42 for lead frames, whose value lies mainly in rolling to tight specifications. Alloy 718 contains about 5 percent niobium. Brazil mines about 93 percent of the world's niobium, and CBMM sells it as vacuum-grade nickel-niobium. Single-crystal turbine blades contain rhenium, of which mines produced only about 81 t in 2025 (CBMM, n.d.; RTX, 2026; U.S. Geological Survey, 2026a, 2026h). These supply facts frame the industry demand in chapter 5.

5. Industry Needs: Demand, Geography and Market Size

Each industry section starts with market size and geography and then covers applications and product forms. The first subsection ends with the best measure at a named stage, and chapter 7 uses a team index where tonnage is missing (indicators in appendix A5). To avoid double counting, electrolyzers and fuel cells sit under energy, process vessels under chemical processing, and sensors under electronics.

5.1 Automotive

Electrification cuts demand for the heat-resistant nickel alloys of combustion vehicles, because electric cars need no turbochargers, exhaust sensors, or glow plugs.

5.1.1 Demand, Geography and Market Size

The world produced 96.4 million motor vehicles in 2025, 34.5 million of them in China. EU car registrations reached 10.8 million, 17 percent below the 2019 level (European Automobile Manufacturers' Association, 2026a; International Organization of Motor Vehicle Manufacturers, 2026).

Over 20 million electric cars were sold in 2025, about a quarter of the market. The IEA expects about half of global sales by 2035 and more than 90 percent in China and the EU (International Energy Agency, 2026d). Batteries do not replace the lost engine alloy nickel, because lithium iron phosphate cathodes contain none and held over 55 percent of the 2025 market (International Energy Agency, 2026d).

European policy may slow the shift. In October 2026, two European Parliament committees backed a 95 percent cut in new car CO2 emissions by 2035, or 87.5 percent with low-carbon steel credits, ahead of a December plenary vote (electrive, 2026).

The best nickel measure covers battery cathodes, at 439 kt in 2025 and 1,294 kt to 1,536 kt in 2035, while declining engine alloy tonnage is not published (International Energy Agency, 2026c).

5.1.2 Applications and Opportunities

Hybrid and gasoline engines keep turbochargers in production for another decade, after close to 50 million units in 2025. Garrett expects output to fall toward its 2022 level by 2030 and the turbo share of gasoline engines to peak at 54 percent in 2032 (Garrett Motion, 2026). BorgWarner won three gasoline turbocharger programs in October 2026, starting in 2027 (BorgWarner, 2026).

Table 3: High performance metal applications in automotive, outlook to 2035

Outlook	Applications	Materials and product forms
Grows	Current sensors and magnetic shielding in electric drives and chargers	Soft magnetic FeNi strip
Stable to 2030	Turbocharger parts for hybrids and gasoline engines	Heat resistant nickel alloy bar and strip
Falls	Glow plugs, diesel exhaust parts, valves and injectors	Nickel alloys, heat resistant stainless strip

Sources: team assessment (BorgWarner, 2026; Garrett Motion, 2026; International Energy Agency, 2026d).

Implications for suppliers:

- Treat engine alloys for cars as a cash business to 2030 and avoid new capacity for diesel only parts.
- Shift product development toward soft magnetic FeNi strip for sensors and shielding in electric drives.

5.2 Oil and Gas

Oil and gas is a stable market for high-performance metals, as investment moves from oil to gas. Sour gas, deepwater fields, and liquefied natural gas (LNG) need alloys that ordinary steel cannot replace.

5.2.1 Demand, Geography and Market Size

Global upstream oil and gas investment fell from USD 625 billion in 2019 to USD 542 billion in 2025 and was about USD 546 billion in 2026 (International Energy Agency, 2026i, 2026k).

In 2025, about 40 percent of upstream spending went to slowing the decline of existing fields (International Energy Agency, 2025b).

North America remained the largest region, at USD 165 billion in 2026 against USD 219 billion in 2019. Middle East investment rose from USD 75 billion to USD 93 billion, with gas roughly doubling to USD 45.4 billion, and Central and South America grew fastest, from USD 42 billion to USD 68 billion (International Energy Agency, 2026k). Under current policies, LNG trade could rise from 560 bcm in 2024 to 880 bcm in 2035 (International Energy Agency, 2025c, 2026k). Alloy and clad pipe tonnes are not published, so investment is the best measure.

5.2.2 Applications and Opportunities

Customer approval is the main entry barrier. ISO 15156 sets which alloys suit sour service, but producers still need NORSOK M-650 qualification and operator approval for each product form (International Organization for Standardization, 2020; Standards Norway, 2011). Orders flow through the vendor lists of EPC contractors and subsea firms such as SLB and TechnipFMC, which booked USD 10.1 billion in subsea orders in 2025 (SLB Limited, 2026; TechnipFMC, 2026).

Forms include nickel alloy bar for wellheads and completion tools, corrosion-resistant tubing, umbilical tube, and clad pipe. Alleima's hyper duplex SAF 3007 succeeds super duplex SAF 2507 in umbilicals, and clad pipe maker BUTTING reports EUR 880 million in turnover (Alleima, n.d.-b; BUTTING, 2026).

LNG carriers add a smaller FeNi market than trade growth suggests. Only GTT's NO96 membrane uses Invar, a 36 percent nickel steel, in 0.7 mm strakes, while Mark III and Mark III Flex use corrugated 304L stainless steel (Gaztransport et Technigaz, 2026a, 2026b). FeNi36 demand therefore follows the unpublished NO96 share of orders. Aperam doubled Imphy capacity in 2025 to supply shipyards with Invar M93 (Aperam, 2025, n.d.).

Table 4: High performance metal applications in oil and gas, outlook to 2035

Outlook	Applications	Materials and product forms
Grows	Clad and lined pipe, subsea equipment	Nickel alloy strip, plate, welding wire
Option	NO96 membrane tanks on LNG carriers	FeNi36 strip
Stable	Well tools, wellheads, well tubing, valves and fittings for sour gas	Nickel alloy bar, billet, plate, corrosion resistant tubing

Sources: team assessment (Alleima, n.d.-b; BUTTING, 2026; Gaztransport et Technigaz, 2026b; TechnipFMC, 2026).

Implications for suppliers:

- Monitor sour service: keep NORSOK M-650 and operator approvals current and serve vendor lists from existing capacity.
- Sell strip and welding wire to clad pipe makers selectively, and keep NO96 FeNi36 strip as an option.

5.3 Energy and Environmental Technology

Energy technology shows two trends: gas turbine makers struggle to fill orders, while many hydrogen projects have yet to place any.

5.3.1 Demand, Geography and Market Size

Gas turbine orders grow as data centres need power and grids need stability. Siemens Energy is sold out until fiscal 2028 and plans to raise annual gas turbine capacity from about 17 GW in fiscal 2024 to more than 30 GW (Siemens Energy, 2025). Gas-fired power investment rose from USD 64 billion in 2019 to USD 112 billion in 2025, and data centres could use about 945 TWh in 2030, twice their 2024 use (International Energy Agency, 2025a, 2026k).

Hydrogen demand for alloys moves far more slowly. Low-emissions hydrogen output was about 1 Mt in 2025, against total demand above 100 Mt, and the 2030 pipeline fell to 27 Mt (International Energy Agency, 2026e, 2026f). The IEA lists 4.3 GW of operating electrolysis and 19.3 GW at final investment decision or under construction, 60 percent of it nickel-using alkaline units (International Energy Agency, 2026h). Green hydrogen orders at thyssenkrupp nucera fell from EUR 356 million to EUR 26 million in fiscal 2024/25 (thyssenkrupp nucera, 2025). The IEA puts hydrogen nickel use at 2.3 kt in 2025 and 5.2 kt to 32 kt in 2035 (International Energy Agency, 2026c).

Nuclear power offers steadier but slower demand. Of 81 reactors under construction, 40 are in China, and an October 2025 US framework covers at least USD 80 billion of AP1000 reactors (K&L Gates, 2025; World Nuclear Association, 2026b). The superalloy tonnage used in gas turbines is not published.

5.3.2 Applications and Opportunities

Table 5 compares when three energy opportunities generate demand: gas turbine parts now, fuel cells later and less clearly. Bloom Energy plans to double solid oxide output to 2 GW a year by the end of 2026 but does not name its interconnect material (Bloom Energy, 2026).

Alleima is reopening its Sandviken mill to raise steam generator tube capacity by about 60 percent. It melts its own alloys there, so the expansion creates little open demand for hollows (Alleima, 2026a).

Table 5: High performance metal applications in energy technology, outlook to 2035

Outlook	Applications	Materials and product forms
Grows strongly	Gas turbine blades, vanes, combustors and repair	Superalloy remelt stock, combustor sheet, welding wire
Grows from 2030	Nuclear steam generator tube	Alloy 690 tube, mostly from integrated mills
Option	Solid oxide cell interconnects, alkaline electrolyser electrodes	Coated ferritic strip, nickel strip and mesh

Sources: team assessment (International Energy Agency, 2026h; Siemens Energy, 2025; thyssenkrupp nucera, 2025; World Nuclear Association, 2026b).

Implications for suppliers:

- Qualify superalloy remelt stock and combustor sheet with turbine casting houses while capacity is sold out to 2028.
- Treat electrolysers, fuel cells and nuclear tube as options released only against signed volumes or outside hollow purchases.

5.4 Aerospace

Aerospace is a major growth source for high-performance metals. Jet engine hot sections use nickel and cobalt superalloys, which suppliers cannot melt and forge fast enough.

5.4.1 Demand, Geography and Market Size

Airbus expects the passenger fleet to grow from 23,310 aircraft in 2025 to 45,550 in 2045, needing 42,060 new aircraft, and Boeing expects 43,625 deliveries on a broader definition (Airbus, 2026a; Boeing, 2026b). Engine output grows faster, and CFM delivered 1,802 LEAP engines in 2025, up 28 percent (GE Aerospace, 2026a; Safran, 2026a).

Maintenance and defence add further demand. Airbus projects aviation services to grow from USD 209 billion in 2026 to USD 422 billion in 2045, with materials above 85 percent of off-wing maintenance cost (Airbus, 2026c). With world military spending at USD 2,887 billion in 2025, GE delivered 635 defence engines, up from 490 (GE Aerospace, 2026a; Stockholm International Peace Research Institute, 2026).

Cobalt is the best measured indicator. The Cobalt Institute estimates that aviation used 11,499 t of the 20 kt of cobalt in superalloys in 2025, and the USGS reports that 51 percent of US cobalt goes into superalloys, mainly for engines. Nickel alloy tonnage is not published (Cobalt Institute, 2026; U.S. Geological Survey, 2026d).

5.4.2 Applications and Opportunities

Rotating parts set the highest entry bar in Table 6. At ATI in Lockport and Special Metals in Huntington, disc and shaft billet passes through vacuum induction melting, electroslag remelting, and vacuum arc remelting. The most demanding discs use powder, whose quality Pratt and Whitney's geared turbofan issue put in focus (ATI, 2026a; RTX, 2026; Special Metals Corporation, n.d.). Engine makers also specify approved melt routes, and GE Aerospace says first approval for a rotating part can take several years (GE Aerospace, 2026a).

Table 6: High performance metal applications in aerospace, outlook to 2035

Outlook	Applications	Materials and product forms
Grows strongly	Discs, shafts, casings and rings for LEAP, GE9X, Trent and military engines	Triple melt superalloy bar and billet, powder for discs, sheet
Grows	Engine repair and overhaul	Welding wire, repair alloys
Grows	Additively manufactured combustor and nozzle parts	CoCr and nickel superalloy powder
Stable	Combustor and wear parts, with nickel alloys replacing some cobalt grades	Cobalt alloy sheet and plate

Sources: team assessment (Airbus, 2026a; ATI, 2026a; Safran, 2026a).

Producers answer shortages with supply agreements and new capacity. ATI earned USD 1,762.9 million from commercial jet engines in 2025 under long-term agreements with GE Aerospace, Safran, Rolls-Royce, and Pratt and Whitney, and reports forging lead times above two years (ATI, 2026a, 2026b). Safran committed EUR 150 million to a 30,000 t press at Gennevilliers for about 14,000 hot section parts a year from 2029 (Safran, 2026b).

Other melters are expanding their capacity too. Austria's voestalpine Böhler holds about EUR 1 billion of aerospace orders over five years, Japan's Proterial sells gas turbine superalloys, and Aperam bought Universal Stainless for EUR 415 million in 2025 (Aperam, 2026; Proterial, n.d.; voestalpine AG, 2026). Cannon-Muskegon, part of PCC, melts nickel and cobalt casting alloys for investment casters in five vacuum furnaces (Cannon-Muskegon, n.d.). In China, Western Superconducting's superalloy unit raised first half 2025 revenue 83 percent to RMB 282 million, and a market report puts national output near 64.5 kt (Western Superconducting Technologies, 2026; Zhiyan Consulting, 2026). A new qualified European supplier can still enter, but that window narrows as this capacity arrives around 2030.

Implications for suppliers:

- Start part level qualifications with engine makers and independent forgers now, because rotating approvals take several years.

- Build the repair wire and powder business alongside, because off wing maintenance grows by 3.7 percent a year (Airbus, 2026c).

5.5 Chemical Processing

Chemical processing has long used corrosion-resistant alloys, but European demand now comes mainly from maintenance while new plants rise in Asia.

5.5.1 Demand, Geography and Market Size

Europe's chemical industry produces less and has spare capacity, with EU27 sales of about EUR 635 billion in 2024. Europe's share of world sales fell from 21 to 13 percent since 2009, while China's rose from 24 to 46 percent (European Chemical Industry Council, 2025a, 2025b). EU chemical output fell 3.2 percent in 2025, and utilization was 75.0 percent in the third quarter of 2026, against a long-term 81.3 percent (European Chemical Industry Council, 2026b; Eurostat, 2026b).

New investment follows that growth to Asia. BASF expects Asia Pacific to provide over 80 percent of chemical growth by 2035 and opened its Zhanjiang Verbund site in November 2025 (BASF, 2026). The closest metal stage measure is EU use of nickel flat products, 57.8 kt in 2024 across all end uses, with an unknown chemical share (Eurostat, 2025).

5.5.2 Applications and Opportunities

Corrosion-resistant alloys are needed where stainless steel fails, as in sulfuric acid, hydrochloric acid, or chlor-alkali service. In Table 7, European demand is mainly replacement, while new plants and clad plate move to Asia. Alleima is investing SEK 220 million in finishing at Mehsana, India, where its range includes Sanicro 625 (UNS N06625) tube of 6 to 42 mm outside diameter (Alleima, 2025, 2026b).

Table 7: High performance metal applications in chemical processing, outlook to 2035

Outlook	Applications	Materials and product forms
Stable	Replacement heat exchangers, vessels and piping in European plants	C-276, C-22 and 59 type NiCrMo alloys and Alloy 625 as plate, sheet and welding consumables
Grows	New plants in China, India and Southeast Asia	Plate, clad plate, zirconium plate and tube
Selective	Battery material plants, where titanium and linings dominate	Plate for selected duties, welding consumables
Falls	New build in European base chemicals	Plate

Sources: team assessment (Alleima, 2026b; BASF, 2026; European Chemical Industry Council, 2026b).

Implications for suppliers:

- Harvest European replacement demand: sell plate and welding consumables from existing capacity on documented life cycle cost.
- Serve Asian EPC tenders only from existing plate capacity, without new investment.

5.6 Electronics and Electrical Engineering

Electronics has the largest headline sales but the smallest metal volumes, often tonnes of specialized strip, foil, and high-purity alloy.

5.6.1 Demand, Geography and Market Size

Semiconductor sales reached USD 795.6 billion in 2025, up 26.2 percent, and WSTS forecasts USD 1.51 trillion in 2026 as memory sales more than triple. Sales value does not translate into metal volume (World Semiconductor Trade Statistics, 2026).

Equipment spending is a closer alloy indicator. SEMI expects 300 mm fab equipment spending to reach USD 149 billion in 2026 and USD 227 billion in 2029, in Taiwan, Korea, China, and increasingly the United States (SEMI, 2026b). Electronics demand for FeNi or high-purity alloy is not published in tonnes.

5.6.2 Applications and Opportunities

NVIDIA says that a 1 MW data centre rack at 54 VDC needs up to 200 kg of copper busbar, and it plans 800 VDC Kyber racks in 2027 with partners including Infineon, Vertiv, Eaton, and Schneider Electric. Vertiv raised its copper hedge to 12,946 t by September 2026 (NVIDIA, 2025a; Vertiv, 2026).

Dai Nippon Printing (DNP) doubled OLED metal mask capacity with an 8th-generation Kurosaki line and expects its invar patents to stay protected (Dai Nippon Printing, 2024a, 2024b). Some panel makers test mask-free deposition. TrendForce names Visionox and TCL CSOT, while Samsung Display and BOE keep fine metal masks on 8.6-generation lines (Dai Nippon Printing, 2024b; TrendForce, 2025).

Table 8: High performance metal applications in electronics, outlook to 2035

Outlook	Applications	Materials and product forms
Grows	OLED fine metal masks for IT panels, at risk from mask free patterning	FeNi36 ultra thin foil

Outlook	Applications	Materials and product forms
Grows	Current sensors and shielding in power electronics and data centres	Soft magnetic FeNi strip
Grows	Fab gas delivery and chamber components	High purity nickel alloy and stainless strip and tube stock
Stable	Glass seals, lead frames, resistance heating	FeNi42 and FeNi29Co18 strip, resistance wire

Sources: team assessment (Dai Nippon Printing, 2024a; Evertiq, 2026; NVIDIA, 2025a; SEMI, 2026b).

TSMC, Applied Materials, and Lam Research filings name no alloys, so suppliers sell through component makers. Infineon's EUR 5 billion Smart Power Fab, opened in Dresden in July 2026, shows that Europe still adds power electronics capacity (Evertiq, 2026).

Implications for suppliers:

- Partner with mask makers to qualify FeNi36 ultra thin foil rather than compete with protected processes.
- Qualify high purity alloy strip and tube with the component suppliers of fab gas systems.

5.7 Medical Technology

Medical technology offers the steadiest growth of the seven industries. Aging populations drive demand for cobalt-chromium, titanium, and zirconium implants, largely independent of business cycles.

5.7.1 Demand, Geography and Market Size

Joint replacements are the most measurable driver of implant metal. In 2023, the OECD average was 198 hip and 156 knee replacements per 100,000 people, after a decade of rising rates (OECD, 2025). Europe's medtech market was about EUR 173 billion in 2025, and the United States held 46.4 percent of 2024 device sales (MedTech Europe, 2025, 2026). Central purchasing in China cut hip and knee prices by about 82 percent (National Healthcare Security Administration, 2022).

Because implant alloy tonnage is not published, procedures are the best measure. The 35 OECD members reporting both recorded about 2.75 million hip and knee replacements in 2023, excluding the United States and Japan (OECD, 2025).

5.7.2 Applications and Opportunities

A few large companies buy most implant metals, so approval matters greatly. Stryker reported USD 9,469 million in orthopaedic revenue in 2025, and the four leading groups more than USD 20 billion in implants (Johnson & Johnson, 2025; Smith & Nephew, 2026; Stryker, 2026). Mill demand still swings quickly, as Carpenter's medical sales fell 21 percent to USD 278.4 million in fiscal 2026, or 24 percent to USD 224.3 million excluding surcharge, as customers cut inventories (Carpenter Technology, 2026).

ASTM standards differ by product form. ASTM F75 covers cast CoCrMo, including knee femoral castings and casting alloy sold as ingot or bar. ASTM F1537 covers wrought CoCrMo bar, rod, and wire, and ASTM F3213 covers powder bed fusion, used for the printed talus implant the FDA approved in 2023 (ASTM International, 2017, 2020, 2023; U.S. Food and Drug Administration, 2023).

Smith & Nephew's OXINIUM, an oxidized Zr2.5Nb femoral component with 94.1 percent survivorship at 20 years, replaces some CoCr. It needs wrought Zr2.5Nb bar to ASTM F2384 from an undisclosed supplier (ASTM International, 2024, n.d.; Smith & Nephew, 2026).

Table 9: High performance metal applications in medical technology, outlook to 2035

Outlook	Applications	Materials and product forms
Grows	Cast knee femoral components	CoCrMo casting alloy, ingot and bar to ASTM F75
Grows	Wrought hip and knee components	CoCrMo bar to ASTM F1537
Grows	Additively manufactured implants	CoCrMo powder to ASTM F3213
Grows	Oxidized zirconium femoral components	Zr2.5Nb bar to ASTM F2384
Falls	Implants for China under central purchasing	CoCr bar

Sources: team assessment (ASTM International, 2023; National Healthcare Security Administration, 2022; Smith & Nephew, 2026).

Implications for suppliers:

- Offer CoCrMo casting alloy to ASTM F75 and bar to ASTM F1537 to implant makers in Europe and the United States.
- Develop CoCrMo powder to ASTM F3213 and Zr2.5Nb bar to ASTM F2384, and pass the supplier audits of the device makers.

6. Global Market Size and Trade

Chapter 5 measured each industry at its own stage. This chapter sizes the global market at the mill product stage and then follows trade routes and policies.

6.1 Global Market Size by Product Family

Six digit customs data separate nickel alloys from pure nickel. In 2024, ten major reporters exported 64.5 kt of alloy bars and rods (HS 750512) at an average USD 40.6 per kg. They also shipped 30.6 kt of alloy wire (750522), 41.3 kt of alloy flat products (750620), and 18.0 kt of alloy tubes (750712) (United Nations Statistics Division, 2026c, 2026e). Alloys make up 97 percent of bar and wire exports by weight, 85 percent of flat and 87 percent of tube exports. The four digit totals in section 6.2 therefore track alloy trade closely (team calculation).

Exports are only a floor, so world output has to be estimated. EU apparent consumption, production plus imports minus exports, was 57.8 kt of flat products and 9.2 kt of tubes in 2024, worth EUR 1.8 to 2.8 bn and EUR 0.24 to 0.38 bn at border unit values. Bar data are confidential (Eurostat, 2025).

Table 10 scales these anchors to world team estimates. Superalloys divide the 20 kt of cobalt used in superalloys in 2025 by an assumed cobalt content. Flat products and tubes divide EU use by an assumed EU share of 12 to 30 percent, and bar applies a multiple of 1.0 to 2.5 to exports. The families overlap, so the rows are never summed (Product Strategy 2035 Team, 2026a).

Table 10: Global market size by nickel alloy product family, 2025 (team estimates, not summable)

Product family	Alloy share of exports, 2024	EU use 2024, unit value (EUR/kg)	World 2025, kt P50 (P10 to P90)	World 2025, EUR bn
Superalloy bar, billet and remelt stock	In bar codes	In bar codes	193 (138 to 279)	4.9 to 9.1
Nickel alloy bar, rod and wire	97%	Confidential, 25.4 to 33.4	131 (102 to 169)	3.3 to 4.4
Flat products (plate, sheet, strip, foil)	85%	57.8 kt, 31.0 to 47.8	282 (224 to 364)	8.7 to 13.5
Tubes and fittings	87%	9.2 kt, 26.1 to 41.1	45 (36 to 58)	1.2 to 1.8
Precision FeNi strip and foil	No separate code	Not reported	Not estimable	Not estimable

Sources: UN Comtrade alloy codes, eight reporters, and Eurostat PRODCOM (Eurostat, 2025; United Nations Statistics Division, 2026c, 2026e). Team estimates at constant 2024 unit values, superalloy premium 1.00 to 1.41, method in appendix A2.4 (Product Strategy 2035 Team, 2026a).

The wide ranges still rank the families: flat products lead in tonnes and value, and superalloys rank second in value. The superalloy range is plausible mainly in its upper half, because the cobalt method likely understates melt from revert (Product Strategy 2035 Team, 2026a). Section 6.2 shows where these products are converted and shipped.

6.2 Conversion Creates the Trade Surplus

Trade data also show where conversion happens. Germany leads the ten reporters in flat product exports, ranks second in bars and wire, and raised its mill product exports from USD 0.968 bn in 2019 to USD 1.418 bn in 2024 (United Nations Statistics Division, 2026b, 2026d, 2026f).

Table 11: Leading exporters of nickel mill products and unwrought nickel, 2024

HS code and product	Leading exporters among the ten reporters (USD million)	Germany net trade (USD million)
7505 Nickel bars, rods, profiles, wire	United States 1,400, Germany 638, France 428, United Kingdom 402, Italy 388 (mirror: Austria 384)	+79
7506 Nickel plate, sheet, strip, foil	Germany 578, United States 482, Japan 274, France 110, China 85 (mirror: Sweden 101)	+414
7507 Nickel tubes and fittings	United States 324, Germany 202, China 195, United Kingdom 168, Italy 141	+105
7502 Unwrought nickel	China 2,068, Indonesia 862, United Kingdom 768 (mirror: Canada 1,236, Norway 1,086, Russia 1,049, Australia 981)	−736

Sources: UN Comtrade, reporters Germany, United States, China, Japan, Korea, India, France, United Kingdom, Italy and Indonesia (United Nations Statistics Division, 2026g). Mirror values are their imports from non reporters, a lower bound (United Nations Statistics Division, 2026a). Net trade is a team calculation.

Germany is a net importer of USD 736 million of raw nickel, yet a net exporter of every mill product in Table 11. EU flat product exports reach EUR 31.0 per kg, against EUR 15.5 for the contained nickel, which is the conversion premium of section 4.1 (European Central Bank, 2025; Eurostat, 2025; United Nations Statistics Division, 2026d; World Bank, 2026a).

Reporter data suggest that China leads unwrought nickel exports, but partner import data show that Canada, Norway, Russia, and Australia, all class 1 sources, each shipped about USD 1 bn to the ten reporters, more than Indonesia reports (United Nations Statistics Division, 2026a).

Austria and Sweden appear as converters like Germany, and section 6.3 follows the routes to their buyers.

Figure 2: German exports of nickel mill products rose by about 47 percent from 2019 to 2024

Source: UN Comtrade, Germany as reporter, exports to the world in current USD (United Nations Statistics Division, 2026b, 2026d, 2026f).

6.3 Corridors and Dependencies

The main trade routes follow the buyers. In 2024, Germany exported about USD 111 to 112 million of nickel bar and wire each to the United Kingdom and the United States, both key engine markets. German plate and strip go mainly to Austria, Italy, China, and the United States, where chemical and electronics fabricators operate (United Nations Statistics Division, 2026b, 2026d). British turbojet and gas turbine exports reached USD 39.8 bn in 2024, including re-exports (United Nations Statistics Division, 2026g). The United States exported the most alloy bar, 26.4 kt at USD 44.2 per kg, against 10.4 kt at USD 34.4 per kg from Germany (United Nations Statistics Division, 2026c). These corridors are exposed to the trade policies described next.

6.4 Trade Policy Reshapes Sourcing and Access

Trade policy can change costs and access quickly. In 2025, the United States raised Section 232 tariffs on steel and aluminium to 50 percent and, from 1 August, taxed the copper content of copper semis at 50 percent (The White House, 2025a). The general US duty on nickel mill products (HS 7505 to 7507) is 2 to 3 percent (United States International Trade Commission, 2026).

Additional US tariffs on EU goods have changed several times since 2025. The EU and US joint statement of 21 August 2025 capped the combined tariff on most EU goods at 15 percent, general duty included (The White House, 2025b). In February 2026, the United States ended the reciprocal tariffs and applied a 10 percent surcharge for 150 days (Executive Office of the President, 2026a, 2026b). Since 24 July 2026, a Section 301 action sets the combined EU rate at 10 percent where the general duty is lower, and its exemption lists could not be checked for HS 7505 to 7507 (Office of the United States Trade Representative, 2026). EU mills supplying US forgers may therefore pay up to 10 percent that US mills such as ATI and Carpenter avoid.

The EU Carbon Border Adjustment Mechanism entered its definitive phase on 1 January 2026. It covers ferrochrome and ferronickel but not the nickel products of HS chapter 75, so it raises input costs without protecting alloy mill products (Ernst & Young, 2025). The EU Critical Raw Materials Act lists nickel, cobalt, and tungsten as strategic or critical (European Parliament and Council, 2024). With the quotas and licences of chapter 4, these measures

favour conversion near customers and several qualified sources per critical input, conditions that chapter 7 carries into the growth scenarios.

7. Growth Scenarios to 2035

This chapter turns the market sizes of chapters 5 and 6 into demand paths to 2035.

7.1 Growth Model and Scenario Assumptions

Tier 1 models demand as D = A × I × S, with activity A, alloy mass per unit I and the share S of units using a high performance alloy. Each factor is a 2035 to 2025 ratio, so the index is 100 × (1 + g)^10 with g the growth rate a year. P10, P50 and P90 come from 10,000 draws, calibrated to 20.0 kt of superalloy cobalt in 2025, with all parameters in Table A1 (Cobalt Institute, 2026).

Energy follows gas turbine output (ratio 1.00 to 2.00, Siemens Energy alone) and nuclear (1.05 to 1.45). Electronics follows SEMI fab capacity, not alloy tonnes. Medical follows hip and knee procedures at 3.0 to 4.1 percent a year. Chemical activity moves −1.0 to 2.5 percent a year. Oil and gas follows upstream investment (0.80 to 1.12) and LNG (1.15 to 1.51). Automotive follows an electric share of 40 to 62 percent in 2035 (European Chemical Industry Council, 2026b; International Energy Agency, 2025c, 2026d; OECD, 2025; SEMI, 2026a; Siemens Energy, 2025).

Aerospace combines the aftermarket at 3.0 to 4.2 percent a year with new engine deliveries. The outlooks expect 2,103 to 2,181 deliveries a year to 2045 against 1,393 in 2025, a ratio of 1.25 to 1.76 by 2035. The 2025 base already holds the LEAP ramp to 1,802 engines, so P50 growth of 3.7 percent a year is not pessimism (Airbus, 2026c; Boeing, 2026b; GE Aerospace, 2026a; Product Strategy 2035 Team, 2026a).

The model haircuts guidance, which tends to run ahead of outcomes. For aerospace, its mode is the mean (0.982) of three points, with a median draw of 0.96. Boeing reached a 737 rate of 42 a month in Q4 2025 against a 2026 plan of 47 (0.894). Airbus runs at about 856 of 870 guided deliveries (0.983), and LEAP beat its guidance (1.068) (Airbus, 2026b; Boeing, 2026a; GE Aerospace, 2026c; Product Strategy 2035 Team, 2026a). The IEA upstream ratio of 0.951 mixes price bases and is excluded for aerospace, but it sets the other haircuts (median 0.94). The backtest in A2.3 widens the ranges (International Energy Agency, 2026k).

7.2 Results by Industry, Product, and Region

Measured in tonnes, superalloy bar, billet and remelt stock adds the most volume. It grows from about 193 kt in 2025 to 248 kt in 2035, worth EUR 6.3 to 11.7 bn at 2024 unit values (Product Strategy 2035 Team, 2026a).

Table 13: Alloy demand 2035 by industry, index 2025 = 100

Industry	Cobalt base 2025, 2035	P10	P50	P90	P50 a year	P50 without haircut
Aerospace	11.5, 15.4 kt Co	121	136	151	3.1%	143
Energy	8.5, 11.0 kt Co	104	132	167	2.8%	140
Electronics	None	112	141	177	3.5%	141
Medical	None	128	136	145	3.2%	136
Chemical	None	88	108	134	0.8%	108
Automotive	None	84	102	125	0.2%	108
Oil and gas	None	78	98	122	−0.2%	105

Source: team calculation (Product Strategy 2035 Team, 2026a).

Figure 3: Electronics has the highest alloy demand index to 2035, and automotive stays flat (index, not volume)

Source: team calculation, band P10 to P90, line P50 (Product Strategy 2035 Team, 2026a).

Electronics has the highest index, but it tracks fab capacity, and its alloy volume is unmeasured. Automotive stays flat because combustion parts fall to 69 while electric vehicle parts rise to 200 (Product Strategy 2035 Team, 2026a).

Table 14: Product families 2035, team estimates, never summed

Product family	2025 kt	2035 kt (P10 to P90)	2035 EUR bn	Index P50
Superalloy bar, billet, remelt stock	193	248 (176 to 366)	6.3 to 11.7	129
Plate, sheet, strip and foil	282	326 (255 to 429)	10.1 to 15.6	116
Nickel alloy bar, rod and wire	131	153 (116 to 200)	3.9 to 5.1	116
Tubes and fittings	45	50 (39 to 66)	1.3 to 2.1	112
Precision FeNi strip and foil	None	None	None	129

Sources: team calculation, P50, methods in A2.4 (Eurostat, 2025; Product Strategy 2035 Team, 2026a).

Supply limits raise costs more than they cap volume. Cobalt mine supply falls 30 percent short of 2035 demand, or 36 percent if the Congo quota holds, but superalloys need only 12 percent of it (Product Strategy 2035 Team, 2026a). Table 15 lists every tier change, and the Chinese tungsten licence halt changes none.

Table 15: Shock cases and every change of priority tier

Shock	Exposure	Tier changes
Congo cobalt quota at CMOC entitlement ratio	Mine gap 54%	Turbine remelt stock 1 to 2 (3.70), AM powder 2 to Monitor
Indonesian ore quota cut by 22 to 71%	Mine gap 21 to 55%	Solid oxide strip Option to Monitor

Shock	Exposure	Tier changes
Severe aerospace, two years at −20%	Aerospace index 114	Alloy 718 bar 1 to 2 (3.45), AM powder 2 to Harvest

Sources: team calculation, base scores in Table 17 (Product Strategy 2035 Team, 2026a).

By region, engine and turbine growth sits in the United States and Western Europe. China builds 40 of the 81 reactors under construction, and Asia takes over 80 percent of chemical growth (BASF, 2026; World Nuclear Association, 2026b). Emerging uses stay in the tier 2 register of Table A3 until their signposts are met (Product Strategy 2035 Team, 2026a).

8. Strategic Recommendations

This chapter turns chapters 4 to 7 into priorities and actions for producers.

8.1 Priority Products, Industries, Regions, and Customers

Table 17 scores each opportunity from 1 to 5 for an established alloy producer. Growth weighs 20 percent, market and supply security 15 percent each, and other criteria 10 percent. Priority 1 starts at 3.75 and priority 2 at 3.00, with rules in Table A4 and lower rows in Table A5 (Product Strategy 2035 Team, 2026a).

Table 17: Priority 1 and 2 opportunities, weighted scores (1 = low, 5 = high attractiveness)

Opportunity	Ev	Market	Growth	Supply	Compet.	Tech.	Customer	Barriers	Diff.	Score	Priority
Turbine remelt stock, sheet	M	4	4	3	2	4	5	5	4	3.85	1
Engine bar, Alloy 718	M	4	4	3	2	5	5	4	4	3.85	1
Engine bar, all grades	M	4	4	2	2	5	5	4	4	3.70	2
Fab gas strip and tube	M	2	5	4	3	4	4	3	3	3.60	2
CoCr implant bar, powder	M	3	4	2	3	4	4	5	3	3.45	2
Alloy 690 bar, hollows	M	2	2	5	3	4	4	3	3	3.15	2
AM powder	M	2	4	2	2	4	4	3	3	3.00	2

Source: team assessment (Product Strategy 2035 Team, 2026a). Ev evidence (M measured), competitive intensity 5 = low, Diff. differentiation.

Turbine remelt stock and sheet scores 3.85. It covers vacuum melted superalloy casting stock and combustor sheet, with cobalt as a minor element. Gas turbine output grows 3.6 percent a year, and Siemens Energy is sold out to FY2028. Buyers are turbine makers and casting

houses in the United States and Europe, which German plate and strip already reaches (Siemens Energy, 2025; United Nations Statistics Division, 2026d). Its weak point is a driver built on one turbine maker.

Alloy 718 engine bar also scores 3.85. Alloy 718 carries about 5 percent niobium, mined 93 percent in Brazil, and at most 1 percent cobalt. New engines grow 3.7 percent a year on a pool of up to EUR 4.0 bn. Forgers for GE Aerospace and Safran buy triple melt billet, and Germany ships about USD 111 m of nickel bar each to the United States and the United Kingdom (Special Metals Corporation, 2007; U.S. Geological Survey, 2026h; United Nations Statistics Division, 2026b). ATI, PCC, Carpenter, Aubert & Duval and Daido hold long term agreements, so entrants win as second sources (ATI, 2026b).

Cobalt rich grades such as Waspaloy cut supply security to 2, so engine bar in all lead grades misses priority 1 by 0.05 (Product Strategy 2035 Team, 2026a). Because one melt shop serves both priority 1 rows, the combined vacuum melt programme is priority 1 under three of four weightings, and the 718 variant under all four (Product Strategy 2035 Team, 2026a).

Electronics enters through fab gas strip and tube, growing 4.2 percent a year on a pool of only EUR 0.11 bn, sold to Applied Materials and Lam Research through gas system makers. Priority 2 also covers CoCr for Stryker and Zimmer Biomet, Alloy 690 for steam generator tube makers, and AM powder (Stryker, 2026; Zimmer Biomet, 2026). FeNi36 mask foil for DNP and BOE and FeNi36 strip for GTT NO96 tanks stay options until volumes are measured (Dai Nippon Printing, 2024a; Gaztransport et Technigaz, 2026b).

8.2 Recommended Actions and Implementation Roadmap

Producers should qualify first and add capacity after customers sign. For 718 billet, VAR and ESR remelting, billet conversion and ultrasonic inspection are the bottleneck, not VIM.

Table 18: Implementation roadmap 2026 to 2035, owners, benchmarks and gates

Initiative, owner	2026 to 2028	2029 to 2035	Benchmark, gate and KPI
1. Vacuum melt programme, quality and metallurgy, board	Qualify 718 billet with two forgers, remelt stock with one caster	Add VAR, ESR, conversion and inspection	Alleima fourth VAR, ATI Monroe VIM 9 kt, capex not published. Gate: 1.5 kt a year signed (EUR 38 to 71 m) by Q4 2027
2. Fab gas strip and tube, quality	Qualify with gas system makers	Supply from existing rolling	Avesta flat about EUR 11,500 per annual tonne. Gate: first order by 2028
3. CoCr implants, quality and regulatory	Keep ASTM F75 and F1537	Add powder after orders	Capex not published. Gate: two agreements by 2028

Initiative, owner	2026 to 2028	2029 to 2035	Benchmark, gate and KPI
4. Alloy 690, nuclear quality	Hold ASME and RCC-M approvals	Hollows for AP1000 and AP300	Alleima tube capacity +60%. Gate: reactor order by 2029
5. Options, technology	FeNi36 foil and NO96 strip pilots	Scale after signposts	Aperam Imphy doubled. Gate: approval by 2028
6. Inputs, procurement	Second cobalt and niobium sources	Closed loop revert	Four refiners make 15 kt of alloy cobalt. KPI: no source above half

Sources: company reports, team arithmetic (Alleima, 2026a, n.d.-a; Aperam, 2025; Argus Media, 2026b; ASTM International, 2020, 2023; Cobalt Institute, 2026; Outokumpu, 2026b).

The main risks are slow qualification, ceramic composites, Asian capacity and cobalt quotas, each with a kill criterion in Table A15. Producers should stop investing in diesel only parts, whose index falls to 69, and in new European base chemical plants at 75.0 percent utilisation. They should hold electrolyser capacity until orders are funded, avoid captive zirconium cladding, monitor sour service and harvest chemical plate (European Chemical Industry Council, 2026b; Product Strategy 2035 Team, 2026a).

9. Conclusion

The study asked how supply, trade and industrial use of selected metals shape opportunities through 2035. By volume, superalloy bar, billet and remelt stock leads, growing from 193 kt to 248 kt by 2035, worth EUR 6.3 to 11.7 bn (Product Strategy 2035 Team, 2026a). Value comes from conversion into qualified products, while quotas raise cobalt and tungsten costs more than they cap output.

By index, electronics grows fastest at 141, but its alloy volume is unmeasured, so it yields a priority 2 position in fab gas strip and tube. Priority 1 goes to turbine remelt stock and sheet and to Alloy 718 engine bar for engine and turbine makers in the United States and Europe, best run as one vacuum melt programme. Both depend on cobalt supply and aerospace volumes (Product Strategy 2035 Team, 2026a).

Team weights carry the estimates, since official data stop at mill products. Producers should therefore secure inputs, qualify early with engine and turbine makers, and add remelt capacity after customers sign.

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Appendices: Data, Sources, Calculations, and Assumptions

A1 Definitions and the stage ladder

Mt means million tonnes, kt means thousand tonnes, and t means metric tonnes. GW means gigawatts, and bcm means billion cubic meters. Prices are in current U.S. dollars or euros for the year stated.

The value chain has five separate stages: contained metal in mine output, refined metal or intermediates, alloy melt, mill products, and finished components. Examples of intermediates are nickel pig iron and cobalt hydroxide. Mill products include strip, sheet, plate, bar, and wire. The study reports each stage separately. It also separates contained metal from gross weight, actual output from capacity, and commodity benchmark prices such as the World Bank Pink Sheet from prices for qualified alloy products.

A2 Outlook model, backtest and opportunity scoring

This appendix documents the formulas, parameters, tests and scoring rules behind chapters 7 and 8. Values are team estimates unless a source is listed. Running python3 model/outlook_2035.py from the repository root reproduces every result in chapters 7 and 8, writes data/clean/model_*.csv and research/model/outlook_results.md, and calls model/scoring.py for the opportunity scores (Product Strategy 2035 Team, 2026a).

A2.1 Formulas and simulation settings

The model divides each industry j into components k, for example new engines and the aftermarket in aerospace. The 2035 index is I(j) = 100 × Σ w(k) × A(k) × H(k) × (1 + i(k))^10

× S(k) × exp(10 × σ(j) × ε). A is the activity ratio from 2025 to 2035, H the optimism haircut on drivers taken from official guidance, i the annual trend in alloy mass per unit, S the ratio of the share using high performance alloy and w the component weight. σ carries the backtest error into the range, and ε is a standard normal draw.

No trend shift is added to the growth rate. The drivers are official guidance and scenarios, so adding a backtest trend error on top would count the same optimism twice. σ equals half the root mean square growth error of the backtest series, capped at 3 percentage points a year.

A product family index is the weighted sum of the industry indices, with the weights in Table A1. Cobalt tonnes are the 2025 base times the index and the cobalt content ratio. Paths between 2025 and 2035 grow at a constant compound rate for each component.

The simulation uses 10,000 draws and random seed 20261108, with triangular distributions on the low, mode and high values of Table A1. A Gaussian copula links the activity drivers with a global cycle weight of 0.2 and an industry weight of 0.3. Intensity and share are drawn independently. Component and family weights vary in a Dirichlet distribution with concentration 40, or 8 for the FeNi family because its weights rest on less evidence. These settings are team assumptions, because the backtest windows are short and include the pandemic.

A2.2 Parameters and their sources

Table A1 lists each tier 1 parameter, its range and its evidence. A parameter without a cited source is a team assumption, and research/model/outlook_results.md gives the reason for each one.

Table A1: Tier 1 model parameters, low / mode / high

Industry	Parameter	Low / mode / high	Basis and source
Aerospace	Aftermarket growth a year	3.0 / 3.7 / 4.2%	Sourced (Airbus, 2026d; Boeing, 2026b; International Civil Aviation Organization, n.d.)
Aerospace	New engine output ratio 2035 to 2025, before haircut	1.25 / 1.54 / 1.76	Engine delivery path: average deliveries 2026 to 2045 of 2,103 and 2,181 aircraft against 1,393 in 2025 (Airbus, 2026a, 2026c; Boeing, 2026a, 2026b)

Industry	Parameter	Low / mode / high	Basis and source
Aerospace	New engine output 2026 and 2028 over 2025, evidence only	1.11 / 1.18 / 1.20 and 1.25 / 1.39 / 1.44	LEAP guidance of high teens growth and the 2028 plan of 2,500 to 2,600 engines (GE Aerospace, 2026b, 2026c; Safran, 2026a)
Aerospace	Optimism haircut on 2035 activity	0.80 / 0.98 / 1.07, median 0.96	Mean of three plan against outcome points in Table A2 (Product Strategy 2035 Team, 2026a)
Aerospace	Superalloy mass per engine, trend a year	−1.0 / 0.0 / 1.0%	Team assumption (Airbus, 2026c)
Aerospace	Superalloy share of hot section, ratio	0.90 / 0.98 / 1.03	Team assumption, ceramic composites and TiAl
Aerospace	Weights new engines and aftermarket	0.5 and 0.5	Team assumption (GE Aerospace, 2026a)
Energy	Gas turbine output ratio	1.00 / 1.58 / 2.00	Siemens Energy 17 to about 30 GW, one maker only (International Energy Agency, 2026k; Siemens Energy, 2025, 2026)
Energy	Nuclear fleet and tube ratio	1.05 / 1.20 / 1.45	Derived (American Nuclear Society, 2026; International Atomic Energy Agency, 2026; World Nuclear Association, 2026a)
Energy	Other equipment growth a year	2.0 / 3.0 / 3.6%	Electricity demand proxy (International Energy Agency, 2026a)
Energy	Alloy mass per GW, trend a year	−1.0 / 0.0 / 1.0%	Team assumption
Energy	Alloy share of hardware, ratio	0.97 / 1.00 / 1.03	Team assumption
Energy	Weights turbines, nuclear, other	0.55, 0.25, 0.20	Team assumption

Industry	Parameter	Low / mode / high	Basis and source
Energy, oil and gas, automotive	Optimism haircut on 2035 activity	0.85 / 0.95 / 1.02, median 0.94	IEA investment vintage ratio 0.951, price bases differ (Product Strategy 2035 Team, 2026a)
Electronics	Realisation of SEMI path to 2031	0.85 / 1.00 / 1.05	SEMI index 138 in 2031 (SEMI, 2026a)
Electronics	Capacity growth 2031 to 2035 a year	0.0 / 3.0 / 6.0%	Team assumption (SEMI, 2026a)
Electronics	Components and masks growth a year	0.0 / 2.5 / 6.0%	Team assumption (Dai Nippon Printing, 2024b; ZVEI, 2026)
Electronics	Alloy per unit, trend a year	−1.5 / 0.0 / 1.0%	Team assumption (NVIDIA, 2025a)
Electronics	Alloy share, ratio	0.95 / 1.00 / 1.05	Team assumption (Dai Nippon Printing, 2024b)
Electronics	Weights equipment and components	0.6 and 0.4	Team assumption (SEMI, 2026b)
Medical	Hip and knee procedures growth a year	3.0 / 3.5 / 4.1%	Derived, 30 OECD countries 2013 to 2024 (OECD, 2025)
Medical	Other medical uses growth a year	1.8 / 3.0 / 5.6%	MedTech Europe value growth (MedTech Europe, 2025, 2026)
Medical	Alloy per procedure, trend a year	−0.5 / 0.0 / 0.5%	Team assumption
Medical	CoCr and nickel share, ratio	0.90 / 0.97 / 1.02	Team assumption, oxidised zirconium and titanium
Medical	Weights implants and other	0.8 and 0.2	Team assumption (Stryker, 2026)
Chemical	Activity growth a year	−1.0 / 1.0 / 2.5%	Team assumption (BASF, 2026; European Chemical Industry Council, 2026a, 2026b)

Industry	Parameter	Low / mode / high	Basis and source
Chemical	Alloy per output, trend a year	−0.5 / 0.0 / 0.5%	Team assumption
Chemical	Alloy share, ratio	0.97 / 1.00 / 1.03	Team assumption
Oil and gas	Real upstream investment ratio	0.80 / 1.00 / 1.12	IEA scenarios (International Energy Agency, 2025c, 2026i)
Oil and gas	LNG trade ratio	1.15 / 1.35 / 1.51	IEA, 560 to 880 bcm (International Energy Agency, 2025c, 2026g)
Oil and gas	Alloy per unit, trend a year	−1.5 / −0.5 / 1.0%	Team assumption, clad pipe saves material
Oil and gas	Corrosion resistant alloy share, ratio	0.97 / 1.00 / 1.03	Team assumption
Oil and gas	Weights upstream and LNG	0.7 and 0.3	Team assumption (International Energy Agency, 2026j)
Automotive	New car sales ratio	0.95 / 1.00 / 1.10	Team assumption (European Automobile Manufacturers' Association, 2026a; International Energy Agency, 2025d)
Automotive	Electric share of sales 2035	40 / 50 / 62%	IEA scenarios (International Energy Agency, 2026d)
Automotive	Heat resistant alloy per combustion car, trend a year	0.0 / 1.0 / 2.0%	Team assumption (Garrett Motion, 2026)
Automotive	Nickel alloy share of turbo parts, ratio	0.95 / 1.00 / 1.03	Team assumption
Automotive	Precision alloy per electric car, trend a year	−1.0 / 0.0 / 2.0%	Team assumption
Automotive	Weights thermal and electric vehicle parts	0.75 and 0.25	Team assumption (Garrett Motion, 2026)
All	Nickel content of alloy class, ratio	0.98 / 1.00 / 1.02	Team assumption

Industry	Parameter	Low / mode / high	Basis and source
All	Cobalt content, aerospace and other, ratio	0.92 / 1.00 / 1.04 and 0.90 / 1.00 / 1.03	Team assumption (Mining Weekly, 2026)
Families	Superalloy bar, billet and remelt stock	Aerospace 0.575, energy 0.300, car thermal 0.050, oil and gas 0.040, chemical 0.035	Aerospace weight measured (Cobalt Institute, 2026)
Families	Nickel alloy bar, rod and wire	Chemical 0.25, oil and gas 0.25, energy 0.20, aerospace 0.15, medical, car thermal, electronics 0.05 each	Team assumption (Alleima, 2026a; Haynes International, 2023a)
Families	Plate, sheet, strip and foil	Chemical 0.30, energy 0.20, aerospace 0.20, oil and gas 0.15, car thermal 0.10, electronics 0.05	Team assumption (ATI, 2026a)
Families	Tubes and fittings	Oil and gas 0.35, chemical 0.25, energy 0.25, aerospace, car thermal, medical 0.05 each	Team assumption (Alleima, 2026a; International Atomic Energy Agency, 2026)
Families	Precision FeNi strip and foil	OLED mask foil 0.40, EV sensors and shielding 0.25, lead frames and seals 0.35	Team assumption, fab tools use no FeNi (Dai Nippon Printing, 2024b; TrendForce, 2025)
FeNi	OLED mask foil growth a year	−3.0 / 3.0 / 8.3%	DNP mask sales growth against mask free routes (Dai Nippon Printing, 2024a; Product Strategy 2035 Team, 2026a)
FeNi	Share of lines still using fine metal masks, ratio	0.50 / 0.90 / 1.00	Team assumption (TrendForce, 2025)
FeNi	FeNi share of EV sensor and shielding parts, ratio	0.50 / 0.85 / 1.05	Team assumption, competing FeCo, ferrite and nanocrystalline cores

Industry	Parameter	Low / mode / high	Basis and source
Tonnes	Virgin cobalt content of superalloy melt	4 / 8 / 15%	Team assumption (Cobalt Institute, 2026)
Tonnes	Share of superalloy melt leaving as bar, billet or remelt stock	75 / 85 / 95%	Team assumption (ATI, 2026a)
Tonnes	EU27 share of world nickel mill product use	12 / 20 / 30%	Team assumption (International Nickel Study Group, 2026; worldstainless, 2026)
Tonnes	World bar, rod and wire output over the HS 6 traded floor	1.0 / 1.5 / 2.5	Team assumption (United Nations Statistics Division, 2026c)
Value	Superalloy premium over the EU bar unit value	1.00 / 1.22 / 1.41	Sales per finished tonne of a listed alloy producer, EUR 35.8 per kg (Product Strategy 2035 Team, 2026a)
Shock	Severe aerospace case, share of the lost level recovered by 2035	0.0 / 0.7 / 1.0	Team assumption, backlogs favour recovery (Product Strategy 2035 Team, 2026a)

Sources: model/outlook_2035.py and research/model/outlook_results.md (Product Strategy 2035 Team, 2026a). The calibration in data/clean/model_calibration.csv reproduces the 20.0 kt of superalloy cobalt in 2025 exactly and leaves the IEA nickel and cobalt totals for 2035 with plausible remainders (International Energy Agency, 2026c).

A2.3 Backtest and plan against outcome tests

The backtest fits a log linear trend to each series through 2019 and compares it with outcomes from 2020 to the latest year. All series widen the ranges, while the four plan against outcome points in Table A2 set the optimism haircuts.

Table A2: Backtest of tier 1 drivers and plan against outcome tests

Series	Industry	Basis	Test period	MAPE %	Bias %	Ratio or growth error	Used for
IEA upstream investment	Oil and gas	Real	2020 to 2025	17.3	−7.6	−3.45 pp	Range

Series	Industry	Basis	Test period	MAPE %	Bias %	Ratio or growth error	Used for
IEA gas power investment	Energy	Real	2020 to 2025	36.5	−36.0	−20.68 pp	Range
IEA nuclear investment	Energy	Real	2020 to 2025	50.3	−50.3	−16.42 pp	Range
OECD hip and knee procedures	Medical	Counts	2020 to 2024	9.7	9.2	−0.53 pp	Range
OECD STAN output C30, C26, C27, C20, C29	Several	Nominal	2020 to 2023	7.5 to 14.9	−11.0 to 14.9	−5.65 to 0.31 pp	Range
Eurostat C29 volume, EU27	Automotive	Real	2020 to 2025	23.0	23.0	3.22 pp	Range
Eurostat C29 volume, DE, ES, FR, IT	Automotive	Real	2020 to 2025	17.2 to 49.1	17.2 to 49.1	2.02 to 7.02 pp	Range
Boeing 737 rate in Q4 2025 against the 2026 plan rate	Aerospace	Units	2025 to 2026	10.6	11.9	0.894	Haircut
Airbus 2026 run rate against the 870 guidance	Aerospace	Units	2026, projected	1.7	1.7	0.983	Haircut
LEAP 2026 nine month growth against guidance	Aerospace	Units	2026	6.8	−6.3	1.068	Haircut
IEA WEI 2025 estimate against WEI 2026 outcome	Energy, oil and gas, automotive	Mixed price bases	2025	4.9	5.2	0.951	Haircut

Sources: team calculation on IEA, OECD, Eurostat and company data (Airbus, 2026a, 2026b; Boeing, 2026a; Eurostat, 2026b; GE Aerospace, 2026b, 2026c; International Energy Agency, 2025c, 2026k; Leeham News and Analysis, 2026; OECD, 2025, 2026; Product Strategy 2035 Team, 2026a). Bias is the mean of projection minus outcome over outcome. The growth error is the trend slope minus the realised slope, in percentage points a year.

The Boeing point compares the rate reached in Q4 2025 with the rate planned for 2026, so it measures the gap still to close. Boeing expects to be stable at 47 a month only in 2027, which confirms a delay of about one year (Leeham News and Analysis, 2026). The Airbus point projects the full year from nine months at the 2025 seasonality. The LEAP point is counter evidence, since engines beat their guidance while airframe rates slipped. A full year LEAP ratio of 1.000 to 1.014 instead of 1.068 would lower the mean of the three points to 0.959.

The IEA ratio compares an estimate on a 2024 price basis with an outcome on a 2025 price basis, so it mixes price and volume. It is excluded from the aerospace haircut and is kept only for energy, oil and gas, and automotive, with a deliberately wide range. Adding the backtest trend error as a growth shift on top of guidance would give P50 indices of 155 for energy, 116

for oil and gas, 140 for medical technology and 98 for automotive. That would double count the guidance, so the shift is not used (Product Strategy 2035 Team, 2026a).

A2.4 Product family tonnes, values and shock arithmetic

The family tonnes in Table 14 are team estimates at the mill product stage in gross weight, and they overlap, so they are never summed. Superalloy bar, billet and remelt stock divides the 20.0 kt of cobalt in superalloys by a virgin cobalt content of 4 to 15 percent and multiplies by the 75 to 95 percent that leaves the mill as bar, billet or remelt stock. This gives 193 kt at P50 in 2025, with a P10 to P90 range of 138 to 279 kt (Cobalt Institute, 2026; Product Strategy 2035 Team, 2026a).

Two cross checks bound this figure. ATI's fifth VIM furnace at Monroe adds about 9 kt a year from 2027, 5 percent of the P50 (Argus Media, 2026b). China's reported output of 64.5 kt in 2025, taken from a market report and used only for triangulation, equals 33 percent of the P50 (Zhiyan Consulting, 2026).

The other families start from measured trade or consumption. Long products use the HS 750512 and 750522 alloy only exports of eight reporters in 2024, 80.3 kt or 97 percent of all HS 7505 exports, times 1.0 to 2.5 for output that never crosses a border. Flat products divide the EU27 apparent consumption of 57.8 kt by an EU share of world use of 12 to 30 percent, and tubes do the same with 9.2 kt. The HS 6 alloy only floors of 38.2 kt for flat products (85 percent of HS 7506) and 15.3 kt for tubes (87 percent of HS 7507) sit below both results (Eurostat, 2026c; Product Strategy 2035 Team, 2026a; United Nations Statistics Division, 2026c). FeNi foil is not estimable, because no foil tonnes are published.

Values multiply the P50 tonnes by the 2024 extra EU unit values of EUR 25.4 to 33.4 per kg for bar, 31.0 to 47.8 for flat products and 26.1 to 41.1 for tubes. The superalloy family adds a premium of 1.00 to 1.41 over bar (Eurostat, 2025; Product Strategy 2035 Team, 2026a). The 2025 central ranges are EUR 4.9 to 9.1 bn for superalloys, 3.3 to 4.4 bn for long products, 8.7 to 13.5 bn for flat products and 1.2 to 1.8 bn for tubes. The 2035 values keep the 2024 unit values constant. At superalloy values, the 1.5 kt gate of Table 18 equals EUR 38 to 71 m of annual sales, or 0.8 percent of the 2025 family (Product Strategy 2035 Team, 2026a).

The cost effects behind the shock cases of Table 15 are team arithmetic. At 8 percent virgin cobalt, a 25 percent rise in the cobalt price adds about USD 746 per tonne of superalloy and USD 5,595 per tonne of CoCr. A doubling of the 2025 tungsten concentrate price would add about USD 1,917 per tonne of Alloy C-276. Each USD 1,000 per tonne on LME nickel adds about USD 10 per tonne of alloy for every percentage point of nickel, which the monthly surcharge passes on (ATI, 2026a; Haynes International, 2023b; Product Strategy 2035 Team, 2026a; U.S. Geological Survey, 2026d). The severe aerospace case cuts aerospace output by 20 percent in 2026 and 2027 and recovers 70 percent of the lost level. Its 2035 aerospace

index is 114 (96 to 133), the superalloy family index falls from 129 to 117, and aviation cobalt falls from 15.4 to 13.0 kt (Boeing, 2026a; Product Strategy 2035 Team, 2026a).

A2.5 Tier 2 register and further candidates

Table A3 lists the full tier 2 register and the research workbook candidates that are not yet assessed. Two powder positions sit in tier 1 instead: captive disc powder inside the superalloy family, and open market AM powder, scored in Table 17 (Metal AM, 2025; RTX, 2026).

Table A3: Tier 2 register and further candidates

Opportunity	Alloy and product form	TRL and evidence	Earliest volume	Signpost for tier 1
Electrolysers	Nickel strip, wire and mesh	8 to 9, early signal	2026 to 2028	Orders and investment decisions recover
Solid oxide cells	Coated ferritic strip	9 for fuel cells, early signal	2026 to 2028	Purchased strip contract with a cell maker
New additive hot section parts	Superalloy powder	6 to 8, early signal	2027 to 2030	Part specification with a volume call
800 VDC racks	Copper busbar and strip, less copper	6 to 7, early signal	2027	Eaton, Schneider and Vertiv busbar launches
Small reactors	Zirconium cladding, Alloy 690 tube	7 to 8, early signal	2030 to 2033	Serial tube order after a licence
Hydrogen turbines	Castings, forgings, combustor sheet	5 to 7, speculative	2030 to 2035	Order with a hydrogen supply contract
Fusion magnets	Alloy C-276 tape foil	4 to 6, speculative	2030 to 2035	Magnet order for a pilot plant
sCO2 recuperators	Nickel alloy plate and strip	5 to 6, speculative	2032 to 2035	Pilot above 10 MW
Molten salt reactors	NiMoCr plate and tube	3 to 5, speculative	2035 or later	Construction permit and ASME code case
Superhot geothermal casing	Nickel alloy tube and clad pipe	4 to 6, speculative	2032 to 2035	Well above 400 °C with a published casing alloy
Liquid hydrogen aircraft tanks	Aluminium sheet, austenitic tube	3 to 4, speculative	2038 or later	Airframer programme launch
Concentrating solar receivers	Nickel alloy plate, tube, forgings	Not assessed	Not assessed	Workbook candidate
Stationary PEM fuel cell plates	Coated stainless thin strip	Not assessed	Not assessed	Workbook candidate
Robotic surgery and cardiovascular devices	NiTi and CoNiCrMo fine wire and tube	Not assessed	Not assessed	Workbook candidates
Compact electronics cooling	CuCrZr and copper powder	Not assessed	Not assessed	Workbook candidate

Sources: team register and research workbook, TRL and years are team estimates (NVIDIA, 2025b; Product Strategy 2035 Team, 2026a, 2026b; thyssenkrupp nucera, 2026; World Nuclear Association, 2026b).

A2.6 Scoring rules for the opportunity matrix

Table 17 scores opportunities for a producer that already melts and rolls high performance alloys. Market, growth and supply security are computed from data by model/scoring.py. The other five criteria are team judgement on the anchors of Table A4. The weighted score is the sum of each score times its weight. Rows without measured volume (E) are capped at Option, priority 1 starts at 3.75, priority 2 at 3.00 and Monitor at 2.75 (Product Strategy 2035 Team, 2026a).

Market uses the pool specific to the end use. Rows without a cobalt pool use the family tonnes times an end use weight and EU unit values, capped at 4 because the weight is unmeasured. Engine bar thus uses 193 kt times the aerospace weight of 0.575, about EUR 4.0 bn, an upper bound for wrought bar because cast parts share the weight. Growth uses the most specific driver: the engine delivery path for engine bar (3.7 percent a year), gas turbine output for remelt stock (3.6), the SEMI path for fab gas strip (4.2) and the aftermarket for repair wire (3.1).

Supply security takes the weakest input of the lead grades. A single source rule applies where one country mines at least 90 percent of world output with no control in force. Such an element scores 3 below 10 percent of the alloy and 2 above it. Alloy 718, with 4.75 to 5.50 percent niobium from Brazil (93 percent of world output) and at most 1 percent cobalt, scores 3. Cobalt bearing wrought grades such as Waspaloy, at about 13 percent cobalt under the Congo quota, score 2 (Special Metals Corporation, 2007; U.S. Geological Survey, 2026h; Wikipedia contributors, 2026).

The combined vacuum melt programme treats engine bar and turbine remelt stock as one melt shop, with shares of 0.657 and 0.343. It uses the combined pool, the share weighted growth, the weakest supply input and the share weighted judgement scores (Product Strategy 2035 Team, 2026a). The judgement values of the fab gas, NO96 and repair wire rows are first pass team judgement.

Table A4: Scoring rules, weights and evidence

Criterion, weight	Evidence used	Score 5	Score 3	Score 1
Market attractiveness, 15%	Measured or estimated 2025 pool specific to the end use (chapters 5 and 6, A2.4)	At least 10 kt of contained cobalt in the end use (4 from 5 kt, or from EUR 0.75 bn)	2 to 5 kt of cobalt, EUR 0.30 to 0.75 bn, or a measured company niche (2 below these, or a mixed segment)	No measured or estimable volume
Growth to 2035, 20%	P50 growth a year of the most specific driver, tier 2 by earliest volume	4.0% a year or more (4 from 3.0%)	2.0 to 2.9% a year (2 from 0%), or tier 2 volume by 2028	Below 0%, or tier 2 volume after 2033

Criterion, weight	Evidence used	Score 5	Score 3	Score 1
Supply security, 15%	Weakest input of the lead grades (chapter 4, Table 15)	Nickel, iron or chromium without export restriction (4 for a quota on ore or a form melters do not buy)	Cobalt below 10% under the Congo quota, or a single source element below 10%	Active licence halt (2 for a licence regime or quota element above 10%)
Competitive intensity, 10%	Qualified suppliers and long term agreements (A6)	Demand above qualified capacity, no dominant incumbent	Several qualified suppliers competing on service	Price competition with new Asian capacity (2 for an oligopoly with agreements)
Technology attractiveness, 10%	Process route and how many producers master it	Triple melt, foil below 0.1 mm or similar rare routes	Standard grade on a qualified route	Commodity grade in decline
Customer attractiveness, 10%	Buyer backlog, concentration and pricing (chapter 5)	Named buyers with sold out capacity or growing backlog	Fragmented or cyclical buyers	Collapsing orders or state price cuts
Entry barriers, 10%	Time to approval and protection afterwards (A7)	Passable within about two years and protective thereafter (4 for three to five years)	Programme specific or moderate approvals	Captive supply chain closed to outsiders
Differentiation potential, 10%	Scope for grade, tolerance or service advantage	Customer specific grades or tolerances	Partly standardised products	Interchangeable products

Source: team scoring rules (Product Strategy 2035 Team, 2026a).

Table A5 lists the rows below priority 2 and the weighting sensitivities of every row. The equal weight and 20 percent entry barrier columns test whether a tier depends on the base weights.

Table A5: Lower ranked rows and weighting sensitivities

Opportunity	Ev	Mkt, Gro, Sup	Comp, Tech, Cust, Barr, Diff	Score	Tier	Equal weights	Barriers 20%
FeNi36 strip for NO96 LNG tanks	E	1, 3, 5	3, 4, 3, 3, 4	3.20	Option	3.25	3.18
FeNi36 foil and FeNi strip	E	1, 3, 5	2, 5, 3, 3, 4	3.20	Option	3.25	3.18
Solid oxide cell strip	E	1, 3, 5	3, 4, 2, 3, 4	3.10	Option	3.13	3.09
Sour service bar, strip, wire	M	3, 1, 4	3, 3, 3, 5, 3	2.95	Monitor	3.13	3.18
Alloy C-276 fusion tape foil	E	1, 2, 2	4, 5, 2, 3, 5	2.75	Monitor	3.00	2.78
Superalloy repair wire	E	1, 4, 3	2, 3, 3, 3, 2	2.70	Monitor	2.63	2.73
Chemical plate, welding wire	M	3, 2, 2	2, 3, 3, 3, 3	2.55	Harvest	2.63	2.60
Car exhaust and turbo alloys	M	2, 1, 5	3, 1, 2, 3, 2	2.35	Harvest	2.38	2.42
Memo: turbine remelt stock and sheet	M	4, 4, 3	2, 4, 5, 5, 4	3.85	1	3.88	3.98
Memo: engine bar, Alloy 718	M	4, 4, 3	2, 5, 5, 4, 4	3.85	1	3.88	3.87
Memo: engine bar, all grades	M	4, 4, 2	2, 5, 5, 4, 4	3.70	2	3.75	3.73
Memo: combined vacuum melt programme	M	4, 4, 2	Share weighted	3.70	2	3.75	3.77

Opportunity	Ev	Mkt, Gro, Sup	Comp, Tech, Cust, Barr, Diff	Score	Tier	Equal weights	Barriers 20%
Memo: fab gas strip and tube	M	2, 5, 4	3, 4, 4, 3, 3	3.60	2	3.50	3.53

Source: team calculation (Product Strategy 2035 Team, 2026a). The 20 percent barrier column rescales the other weights proportionally, and an equal cut of the others gives the same tiers. Without the haircut, turbine remelt stock and the 718 variant score 4.05, engine bar 3.90 and the programme 3.90, all priority 1 (Product Strategy 2035 Team, 2026a).

A3 Data sources, product codes and reproducibility

The team collected structured data with the project's fetch script. The script saves raw files, creates clean tables, and records each query's date and URL in a source register with more than 1,000 entries.

The UN Comtrade data cover HS 7502, 7504, 7505, 7506, 7507, 8105, 8109, 7219, 7220, 7223, 7409, 7606, 8311, and 8411 for ten reporting countries from 2019 to 2024. The HS 6 digit alloy codes 750512, 750522, 750620 and 750712 separate nickel alloys from pure nickel for 2023 and 2024 and set the traded floors in A2.4 (Product Strategy 2035 Team, 2026a). Eurostat PRODCOM covers fifteen nickel, cobalt, stainless, copper, and aluminium product codes for the EU and five producer countries. Other sources include USGS Mineral Commodity Summaries 2026, the World Bank Pink Sheet, OECD STAN and ICIO, UNIDO INDSTAT for Germany, and filings from 61 companies.

Table 10 in chapter 6 estimates market values by multiplying EU apparent consumption by the lower and higher of the 2024 unit values for exports and imports outside the EU. The team used these figures because Eurostat production values fall below the value of the nickel contained in the products.

For flat products, 57.8 kt × EUR 31.0 per kg gives EUR 1.79 bn, and 57.8 kt × EUR 47.8 per kg gives EUR 2.76 bn. For tubes, 9.2 kt × EUR 26.1 per kg gives EUR 0.24 bn, and 9.2 kt × EUR 41.1 per kg gives EUR 0.38 bn. The team did not value bars because the production figure is confidential (Eurostat, 2025, 2026c).

The team calculated unit values by dividing trade value by trade quantity in PRODCOM DS-059358. The resulting range reflects the mix of grades traded across the EU border. It does not represent a quoted list price.

A4 Limitations and further research

The study has three main limitations. First, official data stop at the mill product stage. HS 6 digit trade data separate nickel alloys from pure nickel, but PRODCOM production data do not, and no source gives tonnes by alloy grade. A survey of producers or Comext data at the eight digit level could sharpen the estimates.

Second, tier 1 uses published activity drivers, but its alloy intensity, use shares, and product family weights are team assumptions. Nickel alloy tonnes by industry are not published, and the backtest periods are short. The aerospace haircut rests on three points of one vintage, and the gas turbine driver rests on Siemens Energy alone, because GE Vernova and Mitsubishi Power data were not available. The team should rerun the model when the IEA World Energy Outlook 2026, the next Airbus and Boeing outlooks, and new SEMI forecasts become available.

Third, the trade analysis covers ten large reporting countries. A global analysis of mirror data that includes Russia, Sweden, and Austria would give a fuller picture of trade routes.

A5 Sector market indicators

The following tables list the market indicators used in chapter 5. Each table identifies the sources and years of observation.

Table A6: Automotive market indicators, 2025 and 2035

Indicator	2025	2035	Source basis
World motor vehicle production	96.4 million	No official forecast	OICA
Electric share of new car sales, world	25%	About 50%	IEA, both exploratory scenarios
Electric share of car sales, China	Close to 55%	Above 90%	IEA
Electric share of car registrations, EU	26.8% (BEV 17.4% plus PHEV 9.4%, team sum)	Above 90% of light duty sales	ACEA, IEA Stated Policies
Turbocharger production	Close to 50 million units	Back toward 2022 level by 2030	Garrett citing S&P Global Mobility

Sources: (European Automobile Manufacturers' Association, 2026a, 2026b; Garrett Motion, 2026; International Energy Agency, 2026d; International Organization of Motor Vehicle Manufacturers, 2026).

Table A7: Oil and gas indicators, today and 2035

Indicator	Today	2035	Source basis
Upstream oil and gas investment	USD 542 bn (2025), USD 546 bn (2026), 2025 prices	Near today's level (Stated Policies), about 10% higher (Current Policies)	IEA
Oil demand	100 mb/d (2024)	Peak near 102 mb/d around 2030, back near 100 mb/d	IEA Stated Policies
LNG trade	560 bcm (2024)	880 bcm (Current Policies)	IEA
Upstream growth regions 2026	Africa +13.7%, Central and South America +9.9%	Continued shift	IEA

Sources: (International Energy Agency, 2025c, 2026k). Regional growth is a team calculation from the WEI 2026 regional data.

Table A8: Energy technology indicators relevant to high performance metals

Indicator	Latest value	Direction to 2035	Metal and product form
Siemens Energy gas turbine capacity	About 17 GW (FY2024), sold out to FY2028	Above 30 GW a year from FY2028	Superalloy bar, billet and forging stock
Reactors under construction	81 reactors, 88.5 GWe gross, 40 in China	710 GW installed in 2035 (IEA Net Zero)	Zirconium cladding tube, Alloy 690 tube
Installed electrolysis	More than 4 GW (2025)	Committed 2030 output 4.3 Mt of hydrogen	Nickel strip and mesh
Solid oxide fuel cells, Bloom Energy	1 GW a year, doubling to 2 GW by end 2026	Driven by data centre power	Interconnect material not disclosed

Sources: (Bloom Energy, 2026; International Energy Agency, 2025c, 2026f; Siemens Energy, 2025; World Nuclear Association, 2026b).

Table A9: Aerospace market indicators, today and 2035

Indicator	Today	2035	Source basis
Passenger fleet, aircraft of 100 seats or more	23,310 (2025)	About 32,600 (team interpolation)	Airbus GMF 2026

Indicator	Today	2035	Source basis
Commercial fleet, wider definition	About 28,000 (2025)	About 38,000	Boeing CMO 2026
New generation share of fleet	39% (2025)	82%	Airbus
Aviation services market	USD 209 bn (2026)	USD 422 bn in 2045, 3.8% a year	Airbus GSF 2026
Cobalt used in aviation	11,499 t (2025)	Grows with fleet and engine output	Cobalt Institute
World military expenditure	USD 2,887 bn (2025), Europe +14%	Rising in Europe	SIPRI

Sources: (Airbus, 2026a, 2026c; Boeing, 2026b; Cobalt Institute, 2026; Stockholm International Peace Research Institute, 2026). The 2035 passenger fleet is a team estimate on a constant growth path and not an Airbus figure.

Table A10: Chemical industry indicators

Indicator	Value	Implication for alloy demand
EU27 chemical sales, 2024	About EUR 635 bn	Large installed base needing maintenance
EU share of world chemical sales	13% (2024), from 21% (2009)	New build moves out of Europe
EU chemical production index, 2025	Down 3.2%	Fewer debottlenecking projects
EU capacity utilisation, Q3 2026	75.0% against 81.3% average	Closures, not expansions
German utilisation, 2025	72.5%	Weakest home market
Asia Pacific share of growth to 2035	Over 80% (BASF)	Export projects through fabricators

Sources: (BASF, 2026; European Chemical Industry Council, 2025a, 2025b, 2026b; Eurostat, 2026b; Verband der Chemischen Industrie, 2026).

Table A11: Electronics indicators relevant to high performance metals

Indicator	Value	Relevance
World semiconductor sales	USD 795.6 bn (2025), USD 1.51 tn (2026 forecast)	Activity, not metal volume
300 mm fab equipment spending	USD 149 bn (2026), USD 227 bn (2029)	Gas delivery and chamber parts in high purity alloys
Copper busbar in a 1 MW rack at 54 VDC	Up to 200 kg	Copper and copper alloy strip and bar
German electrical and electronics value added	EUR 112.2 bn (2023), real output flat in 2025	Home market for sensor and seal alloys
DNP metal mask capacity	Doubled by the 8th generation Kurosaki line	FeNi36 (invar) ultra thin foil

Sources: (Dai Nippon Printing, 2024a; NVIDIA, 2025a; OECD, 2026; SEMI, 2026b; World Semiconductor Trade Statistics, 2026; ZVEI, 2026).

Table A12: Medical technology indicators

Indicator	Value	Relevance
Hip replacements per 100,000 people, OECD average	198 (2023)	Volume driver for CoCr and zirconium bar
Knee replacements per 100,000 people, OECD average	156 (2023)	Volume driver for CoCr bar and powder
European medtech market	About EUR 173 bn (2025)	Devices and diagnostics, not metal
Stryker orthopaedics revenue	USD 9,469 m (2025)	Largest implant buyer
Zimmer Biomet knees and hips	USD 3,322 m and USD 2,094 m (2025)	Names cobalt chrome as a purchased commodity
China volume based procurement	Implant prices cut by about 82% (2022)	Volume without margin

Sources: (MedTech Europe, 2026; National Healthcare Security Administration, 2022; OECD, 2025; Stryker, 2026; Zimmer Biomet, 2026).

A6 Peer benchmark of specialty metal producers

The peer comparison supports the competition scores in section 8.1. It shows the margins of leading specialty metal producers and how they are changing capacity. Howmet and PCC also cast and forge, so they are customers as well as peers.

Table A13: Peer benchmark of specialty metal producers

Company	Revenue, latest year	Margin	Capacity move
Howmet Aerospace	USD 8,252 m (2025)	24.8% operating	Engine products capacity, USD 453 m capex
Carpenter Technology	USD 3,124 m (FY2026)	22.5% operating	Athens, Alabama expansion, capex USD 355 to 375 m guided
ATI	USD 4,587 m (2025)	14.0% operating	Fifth VIM furnace in Monroe, about 9 kt a year from 2027
PCC including Special Metals	USD 10.8 bn (2025)	Not published	Aerospace revenue up 7.5%
Aubert & Duval	EUR 844 m (2024)	Not published	Owned by Airbus, Safran and Tikehau
Alleima	SEK 18,630 m (2025)	8.3% adjusted EBIT	Fourth VAR furnace at Sandviken
Outokumpu	EUR 5,468 m (2025)	3.1% adjusted EBITDA	About 13 kt high nickel flat at Avesta, about EUR 150 m
Aperam	EUR 6,080 m (2025)	5.6% adjusted EBITDA	New VIM complex at Imphy within EUR 160 m programme
Daido Steel	JPY 578 bn (FY2025, search summary)	Not verified	High alloy melting and forging base, trade press
Nippon Yakin Kogyo	JPY 151 bn (FY2025, search summary)	Not verified	Sales down 12% on weak high nickel demand

Sources: company filings and reports (Alleima, 2026a; Aperam, 2026; Argus Media, 2026b; Armees.com, 2025; ATI, 2026a; Berkshire Hathaway, 2026; Carpenter Technology, 2026; Daido Steel, 2026; Howmet Aerospace, 2026; Nippon Yakin Kogyo, 2026; Outokumpu, 2026a, 2026b). Margins as reported or calculated from reported figures, fiscal years differ. Daido and Nippon Yakin figures were read from search result summaries and are not verified.

A7 Qualification gates by industry

The qualification requirements below affect how quickly a producer can enter each application. They support the entry barrier scores in Table 17 and the gates in Table 18.

Table A14: Qualification gates by industry

Industry	Approval gate	Who approves	Implication
Aerospace	AMS specifications, NADCAP accredited processes, part level approval	Engine makers and forgers	Years per part family, start in 2026
Energy	Casting house and turbine maker approvals. For nuclear tube, ASME SB 163 and RCC-M M4105 product approvals plus a nuclear quality assurance programme	Turbine makers, utilities, tube makers	Nuclear approvals only with a named programme
Oil and gas	NORSOK M-650, API 6ACRA and API 5LD, operator approval lists	Operators, pipe and equipment makers	Approval lists are the moat
Chemical	Pressure equipment rules (VdTÜV, AD 2000, ASME), corrosion test data	Fabricators, plant operators	Life cycle cost evidence wins
Medical	ASTM F75 and F1537 material standards and their ISO equivalents, supplier audits	Implant makers with ISO 13485	Few buyers, long relationships

Sources: team summary of the standards named in chapter 5 (Alleima, n.d.-a; ASTM International, 2020, 2023).

A8 Risks and kill criteria

The roadmap in Table 18 depends on the risks below. Table A15 gives an early signal, a response, and a point at which a producer should stop for each risk.

Table A15: Main risks, responses and kill criteria

Risk	Early signal	Response	Kill criterion
Qualification slower than planned	Part approvals behind schedule	Focus on casting stock, which qualifies faster	Less than 1.5 kt a year signed by Q4 2027, so no remelt step in 2028
Substitution by ceramic matrix composites and titanium aluminide	New engine variants with lower nickel content	Join customer development programmes early	Falling hot section nickel content in new programmes
New Asian superalloy capacity	Japanese and Chinese producers win Western approvals	Speed and service close to European customers	Prices below conversion cost in tenders

Risk	Early signal	Response	Kill criterion
Cobalt, niobium and tungsten supply	Quota and licence changes, Congo quota at the CMOC entitlement ratio	Refiners outside the hydroxide chain, revert recovery, stocks	Single source above half of an input
European demand weakness	Chemical utilisation below 75%, falling engine volumes	Shift sales to Asia and the United States	Contribution margin below plan for two years

Source: team assessment based on chapters 4 to 7 (European Chemical Industry Council, 2026b; Product Strategy 2035 Team, 2026a).